EPA Vehicle Fuel Economy

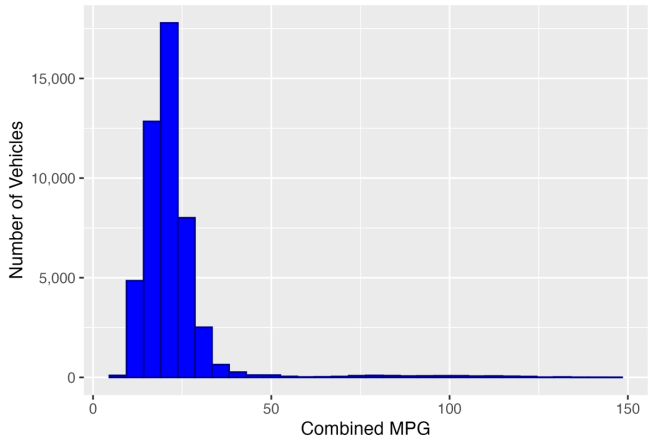
Exploratory analysis and regression model comparison

This case study explores how combined MPG varies across vehicles and compares two simple regression models. Figures come from an original analysis using a vehicle dataset downloaded in November 2024.

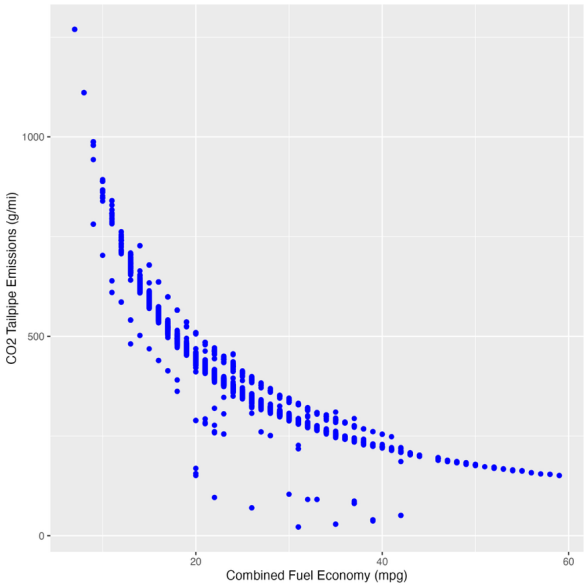
48,202	46,957	1984–2025
Vehicle records	Standard-fuel records	Model-year scope

01 DISTRIBUTION & EMISSIONS

Figures preserved from the original analysis; refreshed runs may differ.



Combined MPG distribution



MPG versus CO2 emissions

Historical interpretation

The full-sample combined MPG mean was 22.08. Its right-skewed distribution included a smaller number of much higher-efficiency vehicles. The emissions scatterplot showed a strong inverse, non-linear association between MPG and tailpipe CO2.

IMPORTANT These findings come from the original report, not a newly run dataset.

From descriptive statistics to models

02 ANALYSIS WORKFLOW

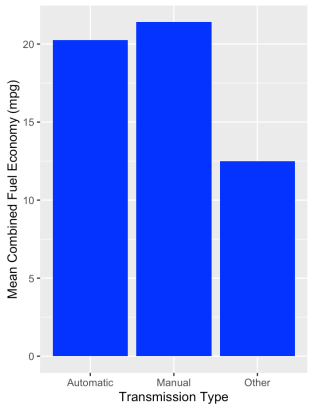
01	**Explore**	Summarize MPG and visualize distributions and CO2.
02	**Filter**	Use non-PHEV records with positive reported tailpipe CO2.
03	**Baseline**	Predict every vehicle with the observed mean MPG.
04	**Compare**	Fit drivetrain and petroleum-consumption regressions.

03 MODEL RESULTS

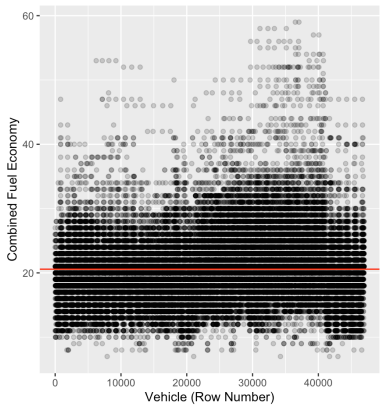
Reported values from the original course analysis:

MODEL	ADJUSTED R^2	RESIDUAL SE
Drivetrain	**0.249**	**4.74 MPG**
Annual petroleum consumption	**0.813**	**2.37 MPG**

04 ORIGINAL CHARTS



Mean MPG by transmission



Intercept-only model

Interpretation matters

Petroleum consumption is calculated from fuel economy. The high R^2 therefore does not establish independent predictive power or causation. The updated scripts also treat ambiguous drivetrain labels conservatively, so newly calculated results may differ from the original table.